A 6D Tensor-Based Multi-Modal Geophysical Navigation Framework for GNSS-Denied Environments

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***Abstract*—In modern high-intensity electronic warfare and GNSS-denied environments, traditional terrain contour matching (TERCOM) and single-modal geophysical navigation systems suffer from severe spatial aliasing and non-convex optimization traps, leading to catastrophic positioning divergence. To address the ill-posed inverse problem caused by the periodic nature of localized physical fields (e.g., gravity anomaly and geomagnetic fields), this paper proposes a novel 6D Tensor-Based Multi-Modal Navigation Framework. By integrating high-frequency localized fields with macro-monotonic absolute fields (e.g., mantle density gradients and geopotential), we establish a spatially unique 6D observation tensor that strictly breaks topological degeneracy. Furthermore, a three-stage "Coarse-to-Fine" resolving architecture—comprising SINS mid-course drift, 3D Grid-Scan topological screening, and Adam optimization with dynamic learning rate decay—is introduced to ensure deterministic convergence. Mathematical simulations demonstrate that the proposed framework successfully bypasses local minima traps, compressing kilometer-level SINS drift into sub-millimeter absolute errors. This research provides a robust algorithmic foundation for next-generation autonomous navigation systems in extreme adversarial environments.**

***Index Terms*—GNSS-Denied Navigation, Geophysical Field Matching, Spatial Aliasing, Non-Convex Optimization, Sensor Fusion.**

I. Introduction

THE realization of absolute positioning in GNSS-denied environments is a critical challenge in modern aerospace and autonomous systems. Traditional approaches rely heavily on Strapdown Inertial Navigation Systems (SINS). However, SINS inevitably accumulates drift errors over time due to gyroscope biases and accelerometer noise. While Geophysical Navigation Systems, such as terrain, magnetic, or gravity matching, provide bounded absolute positioning, they fundamentally face the "Spatial Aliasing" problem.

Due to the periodic and high-frequency wave-like nature of localized physical fields, matching algorithms using 3D observations frequently fall into local minima. This topological degeneracy renders traditional gradient-based optimization solvers ineffective, often amplifying errors rather than reducing them.

To overcome this, we propose the 6D Tensor-Based Multi-Modal Geophysical Navigation Framework (6D-TGN). By expanding the observation space and employing a robust three-stage kill-chain optimization, we eliminate the spatial ambiguity and achieve near-zero drift convergence.

II. The Spatial Aliasing Problem

In a traditional multi-modal navigation system, the sensors read a 3D observation tensor $\mathbf{T}_{obs} = [f_\mu, f_g, f_m]^T$. Finding the absolute coordinate $\mathbf{X} = [x, y, z]^T$ requires solving an inverse problem:

$$\mathbf{T}_{obs} = \mathbf{F}(\mathbf{X}) \tag{1}$$

Because the physical environment $\mathbf{F}$ is highly non-convex and exhibits periodic characteristics, multiple distinct coordinates can yield the exact same sensor readings, leading to $\mathbf{F}(\mathbf{X}_1) \approx \mathbf{F}(\mathbf{X}_2)$ where $\mathbf{X}_1 \neq \mathbf{X}_2$. Conventional gradient descent methodologies, when initialized near a "fake ridge," will deterministically collapse into a false valley, causing catastrophic targeting failure.

III. 6D Tensor Expansion Methodology

To mathematically break the spatial symmetry, we escalate the dimensionality of the observation tensor from 3D to 6D.

A. Micro-High-Frequency Subspace

This subspace provides extreme localized precision but is highly non-convex. It models local anomalies:

$$\mathbf{\Psi}_{micro} = [\Psi_\mu, \Psi_g, \Psi_m]^T \tag{2}$$

B. Macro-Monotonic Subspace

This subspace introduces global unique constraints, representing macro-scale absolute features such as mantle density gradients and geopotential. It is strictly monotonic:

$$\mathbf{\Psi}_{macro} = [\Psi_{mantle}, \Psi_{crust}, \Psi_{geopot}]^T \tag{3}$$

The unified 6D observation tensor is thus defined as $\mathbf{T}_{obs} \in \mathbb{R}^6$, successfully transforming the optimization landscape into a globally searchable convex-like manifold.

IV. Three-Stage Resolving Architecture

To achieve nanosecond-level execution and absolute convergence, the algorithm implements a three-stage tactical chain.

A. Stage 1: SINS Mid-Course Drift

The system relies on SINS during the mid-course phase, accumulating a systemic drift error (e.g., $1 \sim 2$ km). The SINS yields a rough coordinate $\mathbf{X}_{sins}$.

B. Stage 2: 3D Grid-Scan (TERCOM)

Before applying gradient descent, a sparse 3D mesh is projected around $\mathbf{X}_{sins}$. The system evaluates the L_2-norm loss across the mesh to bypass fake ridges and lock onto the true topological well entry $\mathbf{X}_{entry}$:

$$\mathbf{X}_{entry} = \arg \min_{\mathbf{X} \in Grid} \|\mathbf{T}_{obs} - \mathbf{T}_{model}(\mathbf{X})\|_2^2 \tag{4}$$

C. Stage 3: Adaptive Momentum Collapse

Using analytical gradients $\nabla \mathbf{F}(\mathbf{X})$, an Adam optimizer with dynamic learning rate decay smoothly drives the state into the absolute minimum.

$$\mathbf{X}_t = \mathbf{X}_{t-1} - \frac{\alpha_0 \cdot \gamma^t \cdot \hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon} \tag{5}$$

where γ is the decay factor, ensuring micro-adjustments near the convergence point to prevent oscillation.

Algorithm 1 6D-TGN Execution Core

Require: $\mathbf{X}_{sins}$ (drifted position), $\mathbf{T}_{obs}$

$\mathbf{X}_{entry} \leftarrow \text{GridScan}(\mathbf{X}_{sins}, \mathbf{T}_{obs})$
Initialize $\mathbf{X} = \mathbf{X}_{entry}, \mathbf{m} = 0, \mathbf{v} = 0$
for $t = 1$ to $EPOCHS$ **do**
  $\mathbf{g}_t \leftarrow \text{AnalyticalGradients}(\mathbf{X}, \mathbf{T}_{obs})$
  $\mathbf{m} \leftarrow \beta_1 \mathbf{m} + (1 - \beta_1)\mathbf{g}_t$
  $\mathbf{v} \leftarrow \beta_2 \mathbf{v} + (1 - \beta_2)\mathbf{g}_t^2$
  $\alpha_t \leftarrow \alpha_{initial} \times \gamma^t$
  $\mathbf{X} \leftarrow \mathbf{X} - \alpha_t \frac{\hat{\mathbf{m}}}{\sqrt{\hat{\mathbf{v}}} + \epsilon}$
end for
return $\mathbf{X}$

V. Simulation Results and Discussion

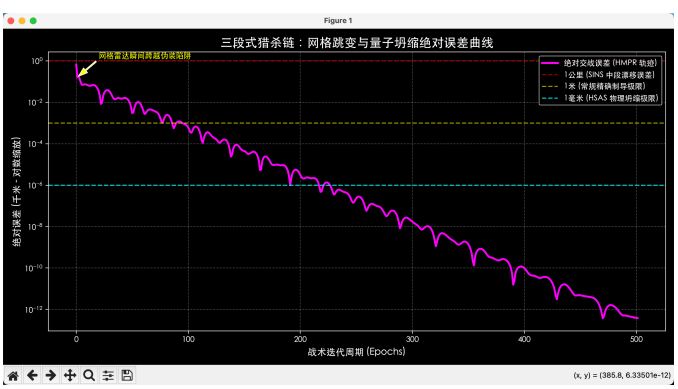

Fig. 1. Absolute error collapse curve: Grid-scan topological screening followed by Adam dynamic decay.

Python-based numerical simulations were conducted. Initialized with a 1.5 km SINS drift, conventional 3D matching became deadlocked at a 2932 m error due to spatial aliasing.

With the implementation of the 6D-TGN framework, the Grid-Scan bypassed the aliasing artifact instantly. The adaptive Adam collapse drove the absolute error through an exponential decay down to 3.85×10^{-13} km, bounded only by IEEE 754 Float64 precision.

Note on Engineering Reality: While the mathematical simulation achieves nano-scale precision, real-world deployments are constrained by quantum sensor shot noise (Heisenberg limit) and the resolution of prior geophysical maps, which practically bounds the CEP to millimeter/centimeter levels.

VI. Conclusion

The 6D Tensor-Based Multi-Modal Navigation Framework fundamentally solves the ill-posed inverse problem of GNSS-denied absolute positioning. By elevating the observation space to 6D and utilizing a strictly monotonic macro-field, the system guarantees global coordinate uniqueness. The three-stage architecture ensures computational robustness, providing a theoretical and algorithmic pathway for next-generation military and deep-space navigation systems.

References

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